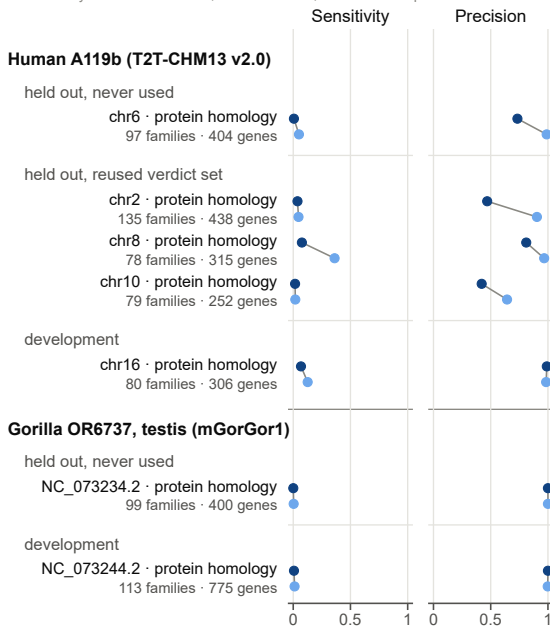


de novo (upper dot and bar)

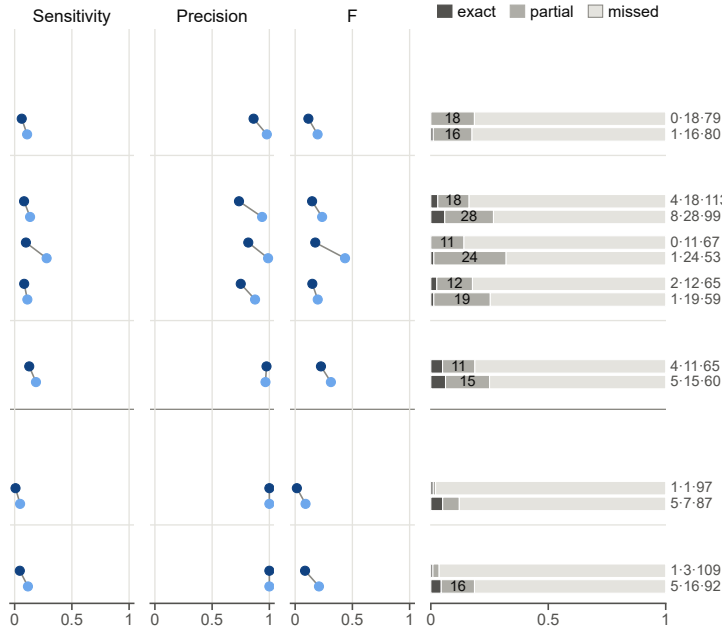
guided (lower dot and bar)

same family rule in both modes; Rustle-internal, not a tool comparison

a Pairwise (gene pairs)



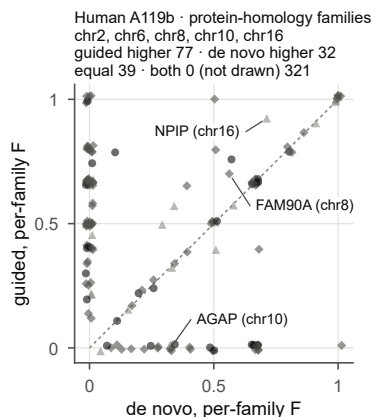
b One-to-one bipartite (family members)



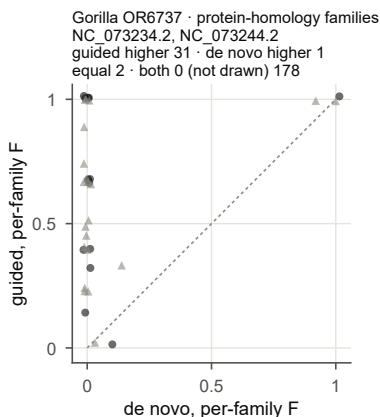
Per-family F: one point per reference family that at least one mode recovers (jitter ≤ 0.015)

● held out, never used ◆ held out, reused verdict set ▲ development

d



e



Supplementary · SECONDARY reference: the annotation's protein-homology families (fold level). Protein is not part of Rustle's family definition. Development tables (per-chromosome runs).